Group Sorting Game

POLI 172 — Games and Human Behavior

Today's Game

Each of you is randomly assigned a **type**: **A** or **B** (50/50 split).

And randomly placed in **Group 1** or **Group 2**.

Your **type** stays the same the whole game. Your **group** can change every round if you choose to switch.

Think of A vs B as two preferences, two candidates, or two cultures.

How a round works

1. The class sees the current composition of both groups.
2. Each group's **outcome** is its majority type. (Ties are broken by a coin flip that round.)
3. You earn **payoff = fit − congestion**.
4. You decide **stay** or **switch** groups for the next round.

Then we repeat.

Payoffs

Outcome in your group	Fit payoff
Your type wins the majority	**+4 points**
Your group is tied	**+2 points**
The opposite type wins	**0 points**

Congestion costs ≈ 2 points per round in a typical evenly-split group. Bigger groups cost more, smaller groups cost less.

Net round payoff = **fit − congestion**. Negative scores are normal.

Two Sessions Today

Session A — about 14 minutes

- 20 rounds
- Switching is **free**

Session B — about 14 minutes

- 20 rounds
- Each switch **costs 1 point**

Between sessions you'll fill out a short reflection survey, then re-enter the same name and email to start Session B.

Game can end early

Each session has up to 20 rounds.

The game ends **as soon as the class is perfectly sorted** — all type A's in one group and all type B's in the other.

This is one of the **Nash equilibria** of the game. Whether and when you reach it is part of what we're testing.

If we sort fast in Session A, you'll see if the switching cost in Session B slows it down.

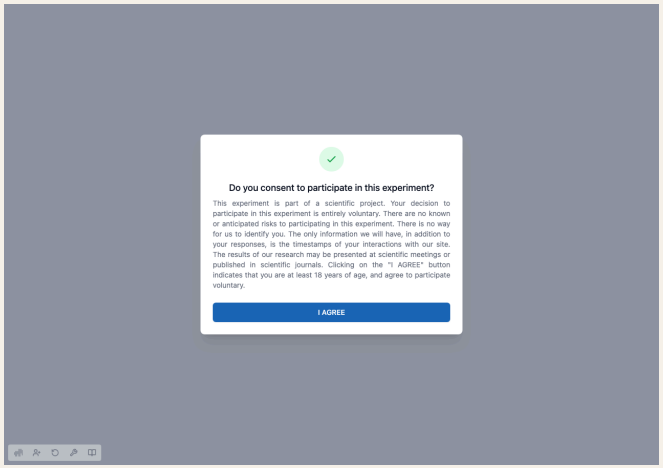
How the experiment works

A walkthrough of what you will see

Consent

First, you will see a consent screen.

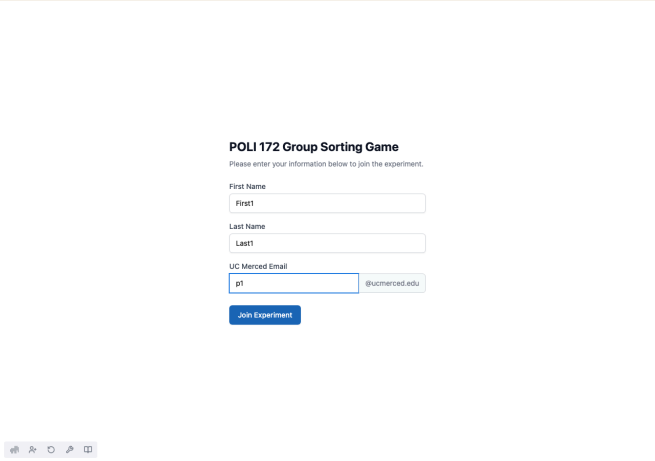
Read it and click **I Agree** to continue.



Step 1: Register

Enter your **first name**, **last name**, and **UC Merced email prefix** (the part before `@ucmerced.edu`).

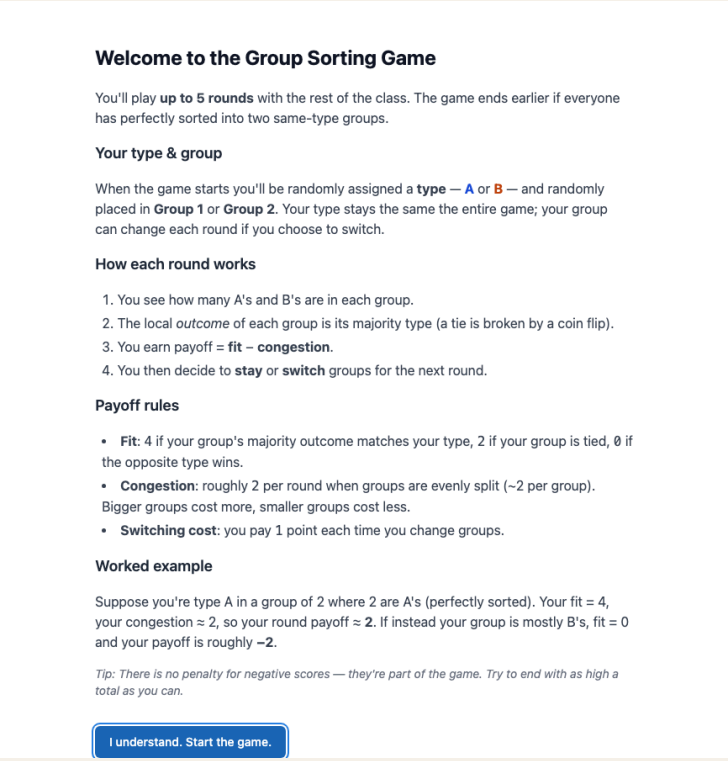
Click **Join Experiment**.



Step 2: Read the instructions

The instructions screen restates the game and shows the actual reward, congestion, and switching-cost numbers for this session.

Read carefully, then click **I understand. Start the game.**



Welcome to the Group Sorting Game

You'll play **up to 5 rounds** with the rest of the class. The game ends earlier if everyone has perfectly sorted into two same-type groups.

Your type & group

When the game starts you'll be randomly assigned a **type** — **A** or **B** — and randomly placed in **Group 1** or **Group 2**. Your type stays the same the entire game; your group can change each round if you choose to switch.

How each round works

1. You see how many A's and B's are in each group.
2. The local *outcome* of each group is its majority type (a tie is broken by a coin flip).
3. You earn payoff = **fit** − **congestion**.
4. You then decide to **stay** or **switch** groups for the next round.

Payoff rules

- **Fit**: 4 if your group's majority outcome matches your type, 2 if your group is tied, **0** if the opposite type wins.
- **Congestion**: roughly 2 per round when groups are evenly split (~2 per group). Bigger groups cost more, smaller groups cost less.
- **Switching cost**: you pay 1 point each time you change groups.

Worked example

Suppose you're type A in a group of 2 where 2 are A's (perfectly sorted). Your fit = 4, your congestion ≈ 2, so your round payoff ≈ **2**. If instead your group is mostly B's, fit = 0 and your payoff is roughly **−2**.

Tip: There is no penalty for negative scores — they're part of the game. Try to end with as high a total as you can.

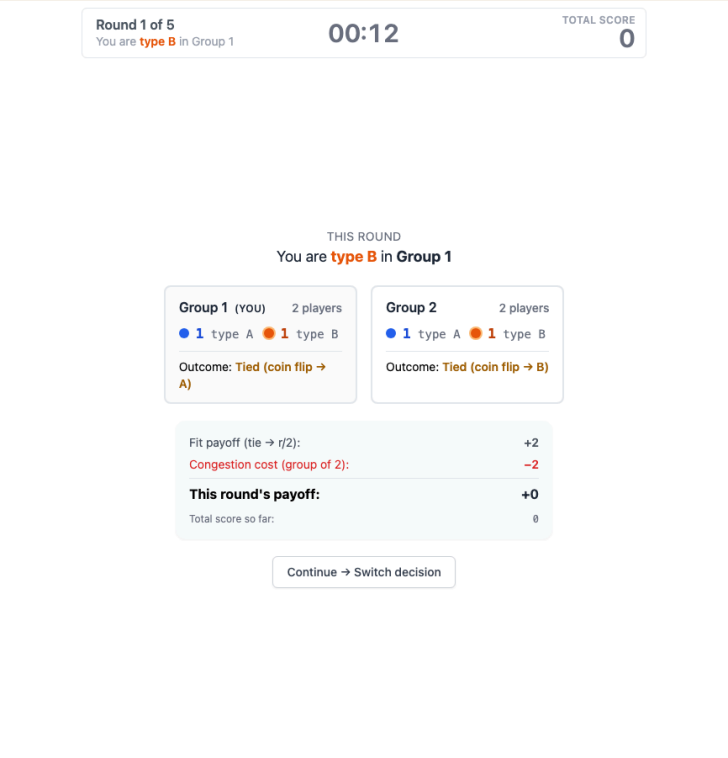
I understand. Start the game.

Step 3: See the round result

Each round shows:

- Your type and current group
- Both groups' A/B counts
- Each group's outcome
- Your payoff for this round (fit, congestion, net)

Click **Continue** to move to the switch decision.

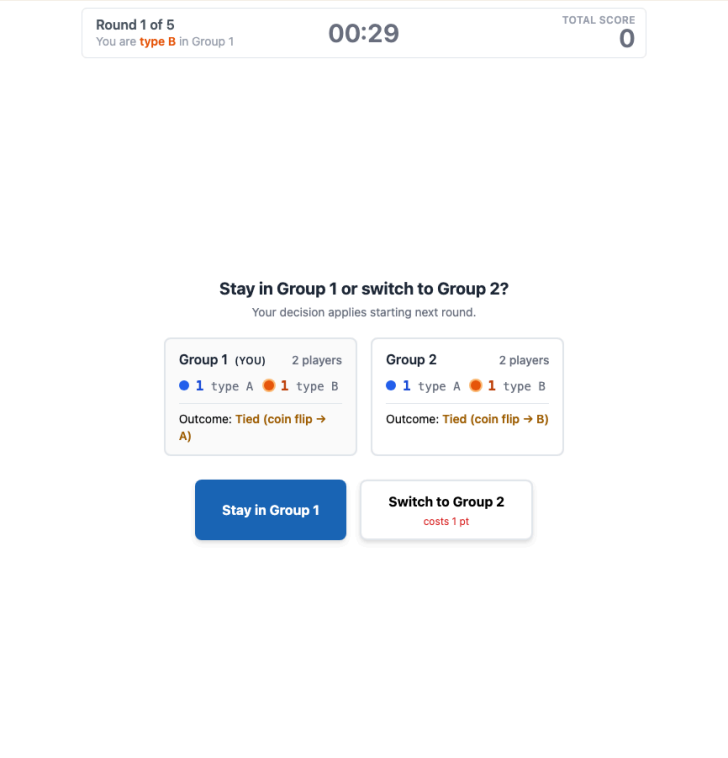


Step 4: Switch decision

Decide to **stay** in your group or **switch** to the other.

Your decision applies starting **next round**.

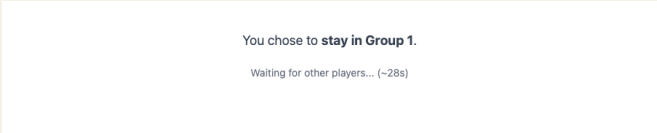
In Session B, the **Switch** button shows “costs 1 pt” — you pay each time you change.



Step 5: Wait for everyone

After you submit, you wait briefly for everyone else to decide.

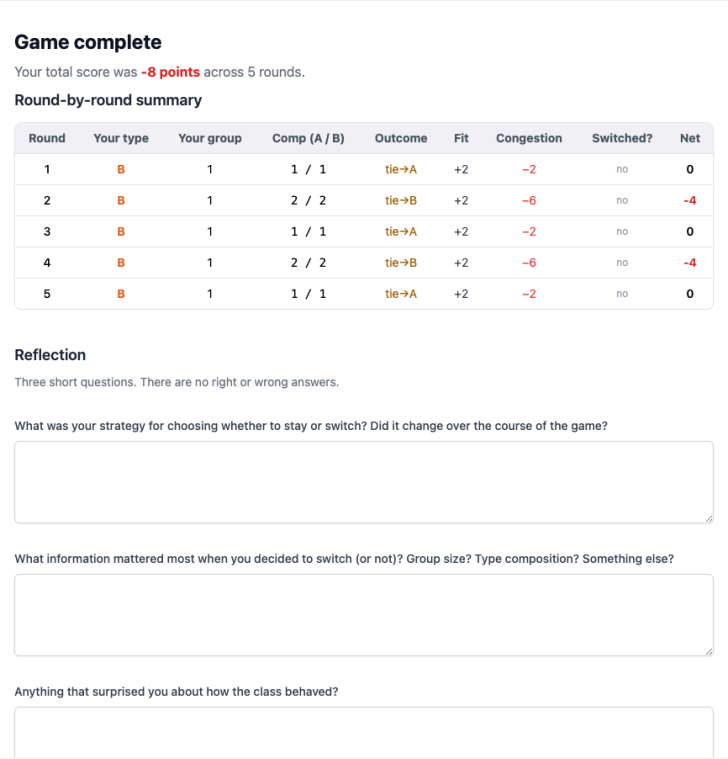
Once all decisions are in, the next round starts.



Exit survey: round summary

After the last round you see a table of every round you played: your type, your group, the composition, fit, congestion, switches, and net payoff.

Use this to reflect on what you tried.



Game complete

Your total score was **-8 points** across 5 rounds.

Round-by-round summary

Round	Your type	Your group	Comp (A / B)	Outcome	Fit	Congestion	Switched?	Net
1	B	1	1 / 1	tie→A	+2	−2	no	0
2	B	1	2 / 2	tie→B	+2	−6	no	-4
3	B	1	1 / 1	tie→A	+2	−2	no	0
4	B	1	2 / 2	tie→B	+2	−6	no	-4
5	B	1	1 / 1	tie→A	+2	−2	no	0

Reflection

Three short questions. There are no right or wrong answers.

What was your strategy for choosing whether to stay or switch? Did it change over the course of the game?

What information mattered most when you decided to switch (or not)? Group size? Type composition? Something else?

Anything that surprised you about how the class behaved?

Exit survey: reflection

Below the round table, **three short questions**:

- What was your strategy for staying or switching?
- What information mattered most when you decided?
- Anything that surprised you about how the class behaved?

Answer thoughtfully — there are no right or wrong answers. Click **Submit** and you're done with that session.

Let’s get started

Open your browser and go to:

poli172.games

Enter your first/last name and UC Merced email prefix, then click **Join Experiment**.

Works on phones, tablets, and laptops. Use any browser.